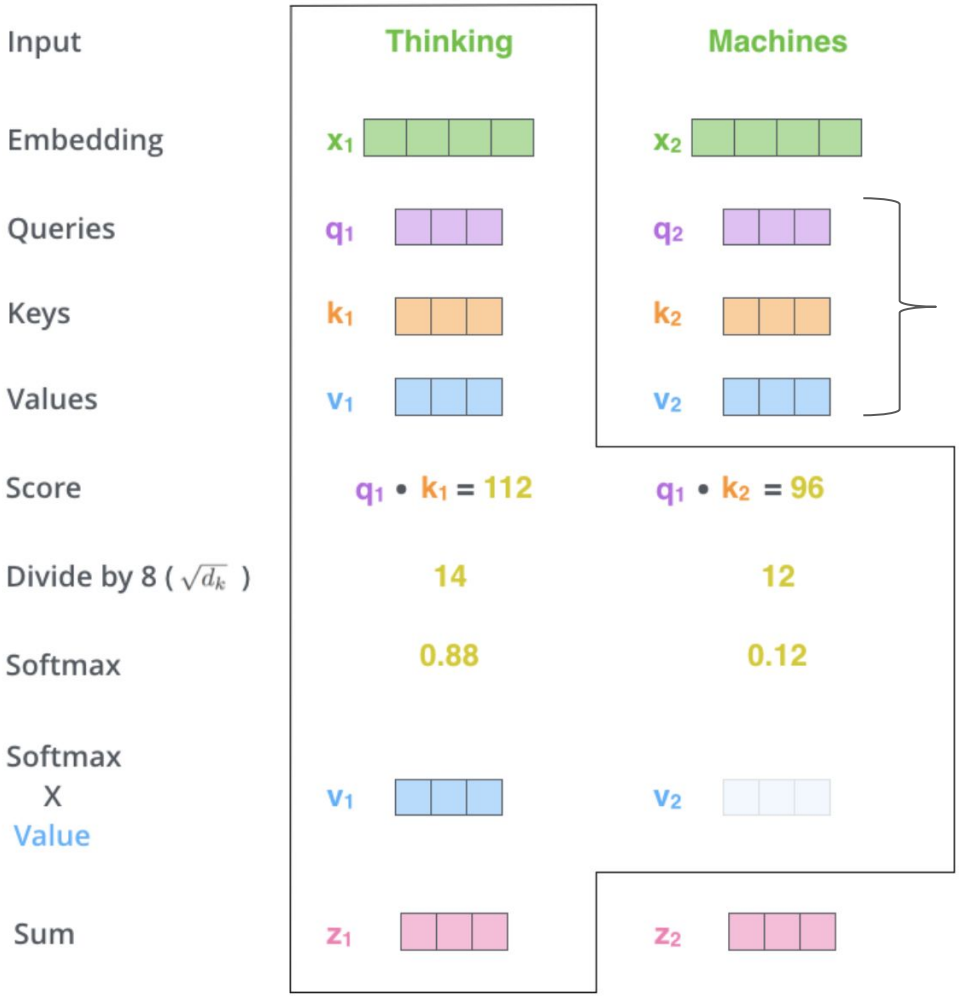
Lecture 05: Transfer Learning, BERT, GPT

Self-Attention: recap

Self-Attention in Detail



STEP 1: create Query, Key, Value

STEP 2: calculate scores

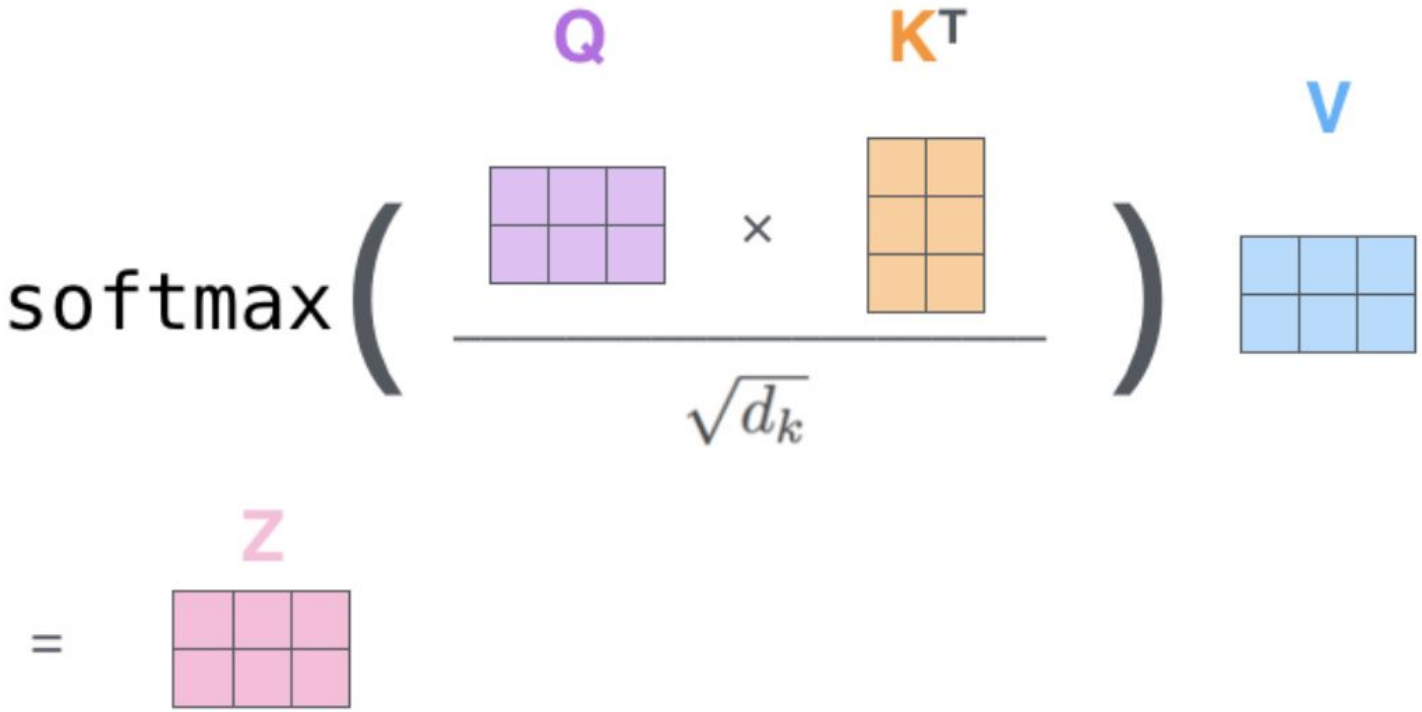
STEP 3: divide by $\sqrt{d_k}$

STEP 4: softmax

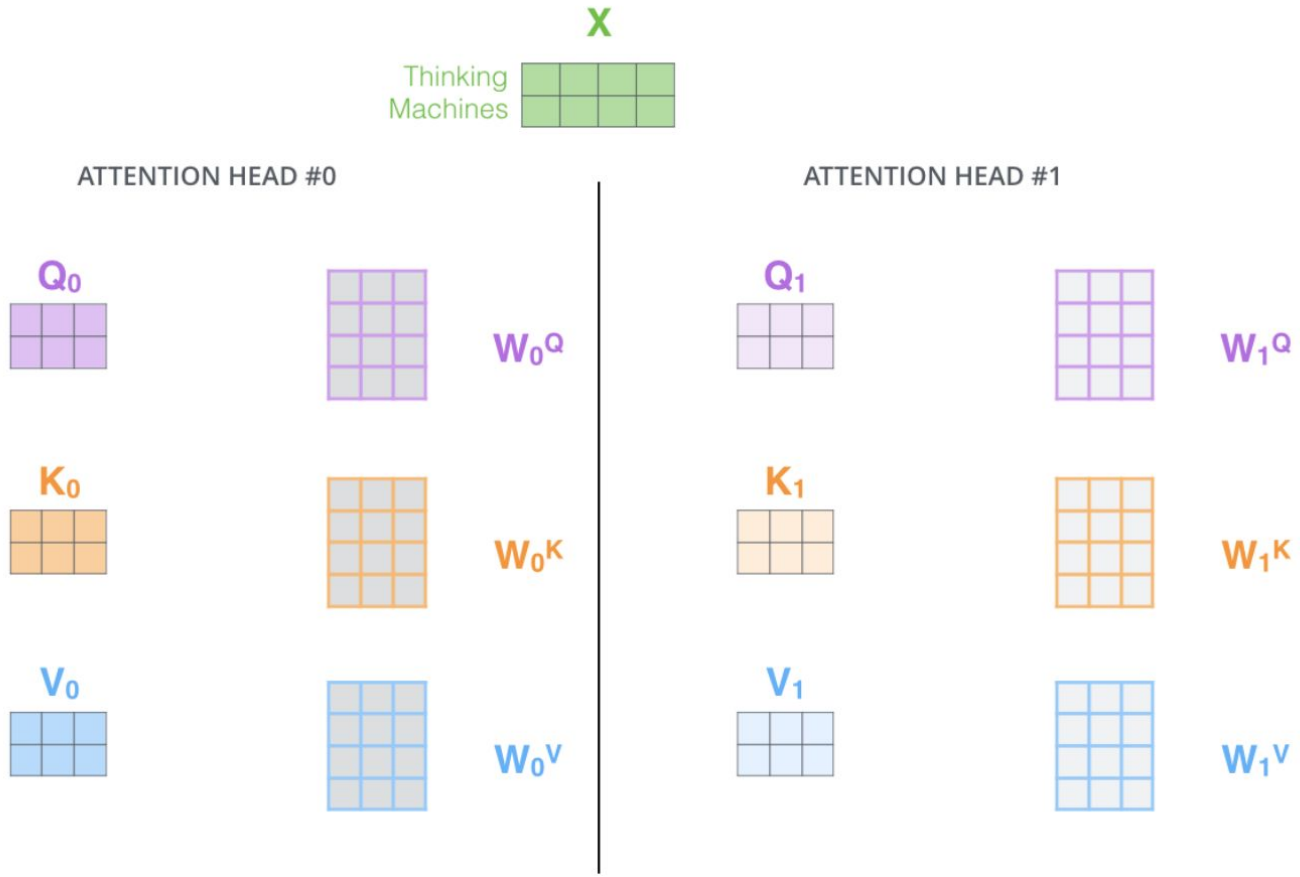
STEP 5: multiply each value vector by the softmax score

STEP 6: sum up the weighted value vectors

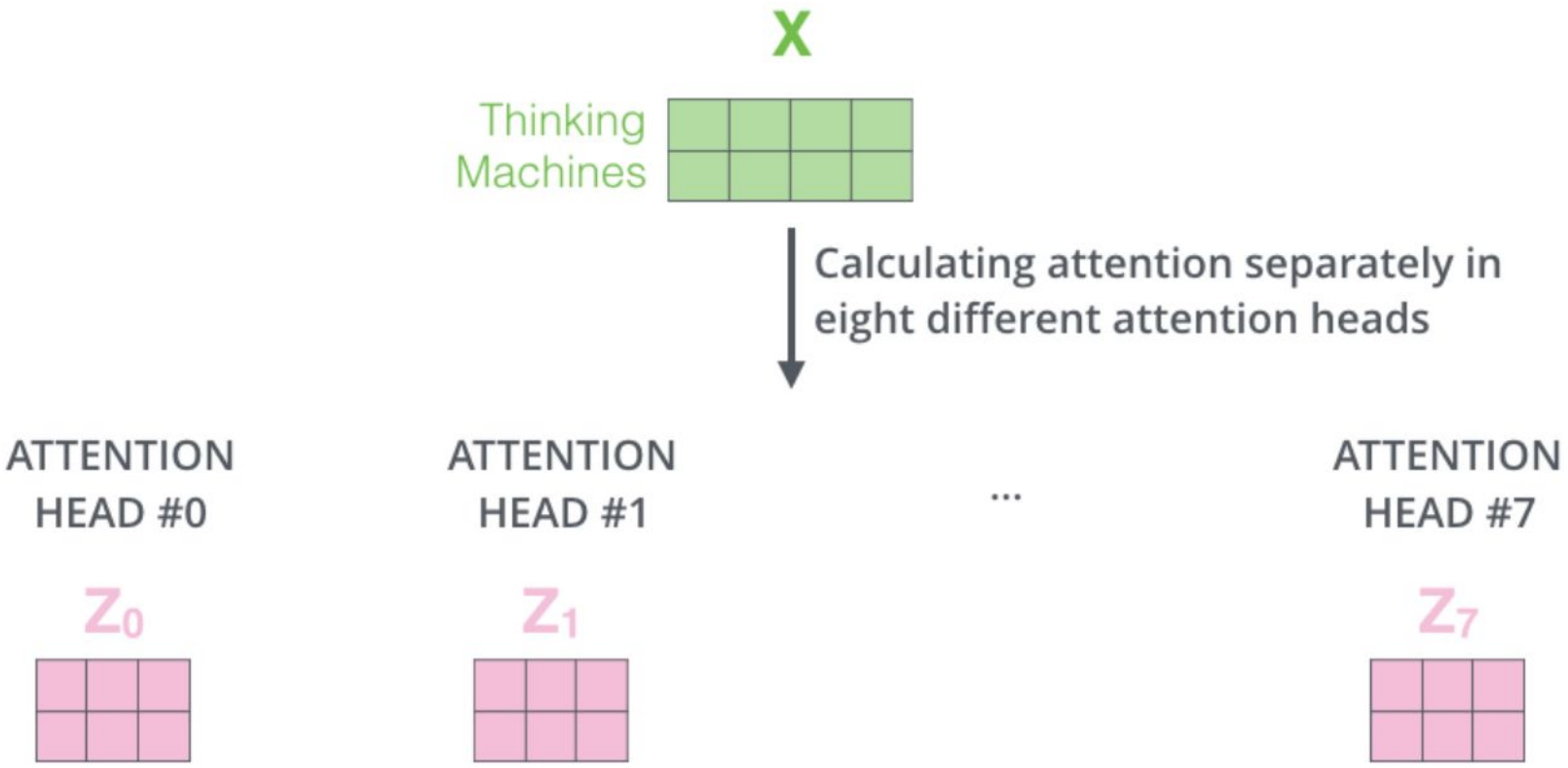
Self-Attention: Matrix Calculation



Multi-Head Attention



Multi-Head Attention



Multi-Head Attention

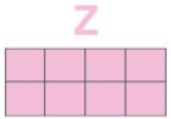
1) Concatenate all the attention heads

2) Multiply with a weight matrix W^O that was trained jointly with the model

X

3) The result would be the Z matrix that captures information from all the attention heads. We can send this forward to the FFNN

=



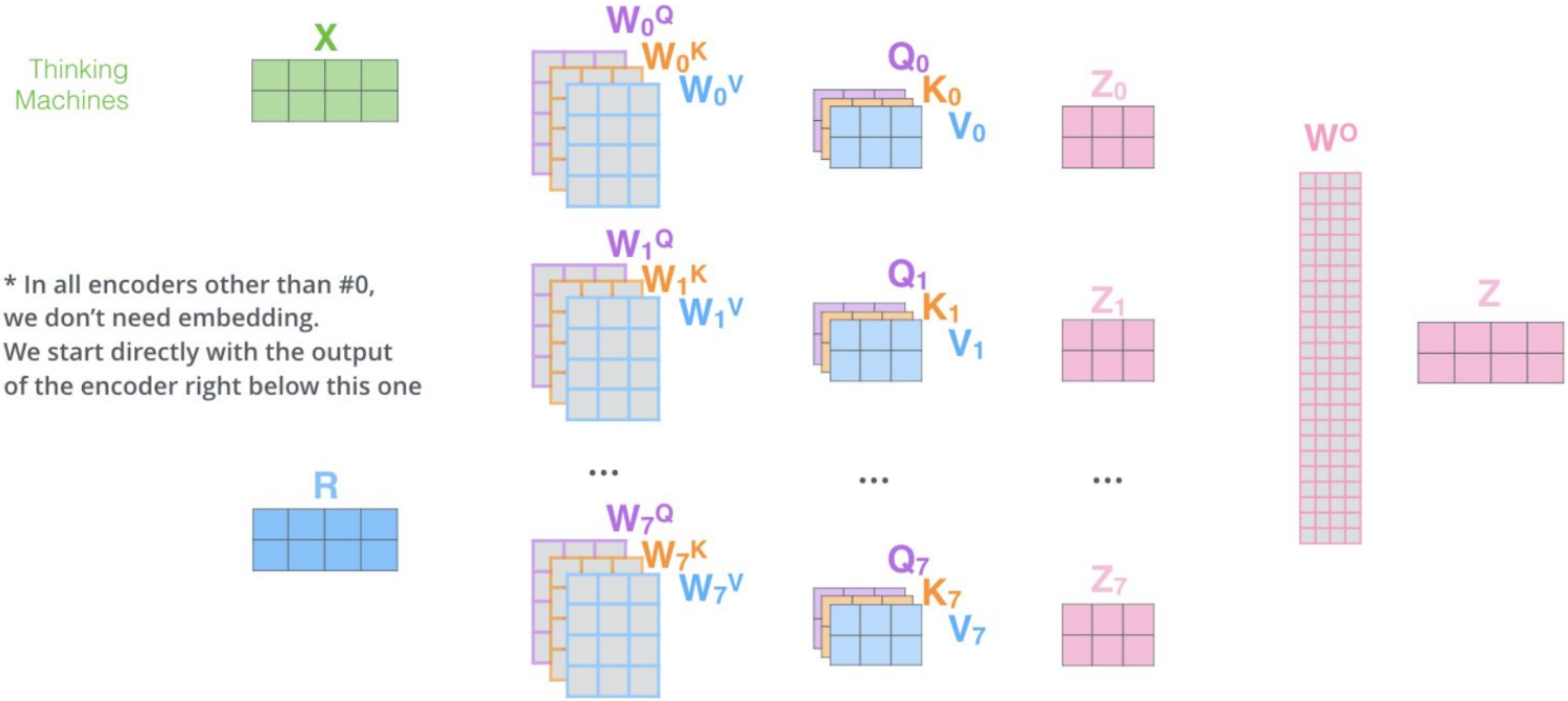
1) This is our input sentence*

2) We embed each word*

3) Split into 8 heads. We multiply X or R with weight matrices

4) Calculate attention using the resulting Q/K/V matrices

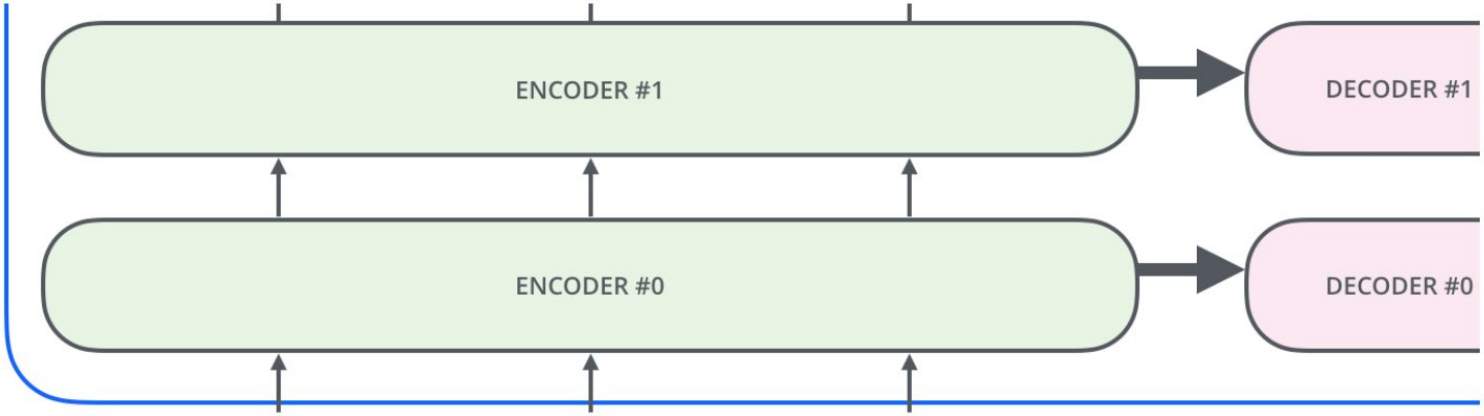
5) Concatenate the resulting Z matrices, then multiply with weight matrix W^O to produce the output of the layer

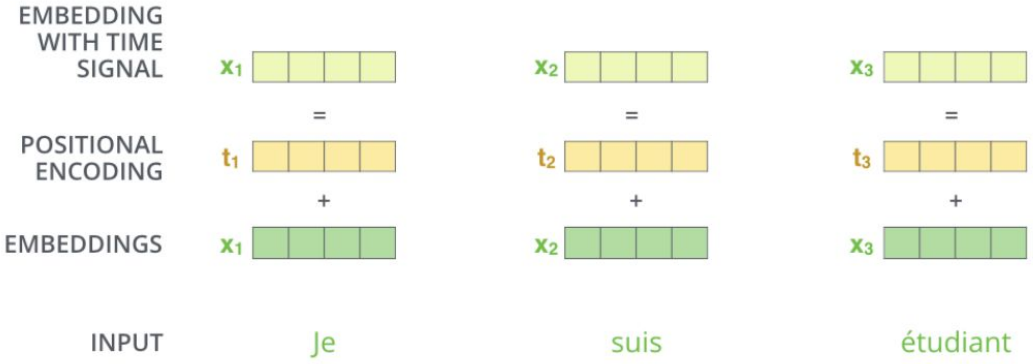


* In all encoders other than #0, we don't need embedding. We start directly with the output of the encoder right below this one

Positional Encoding

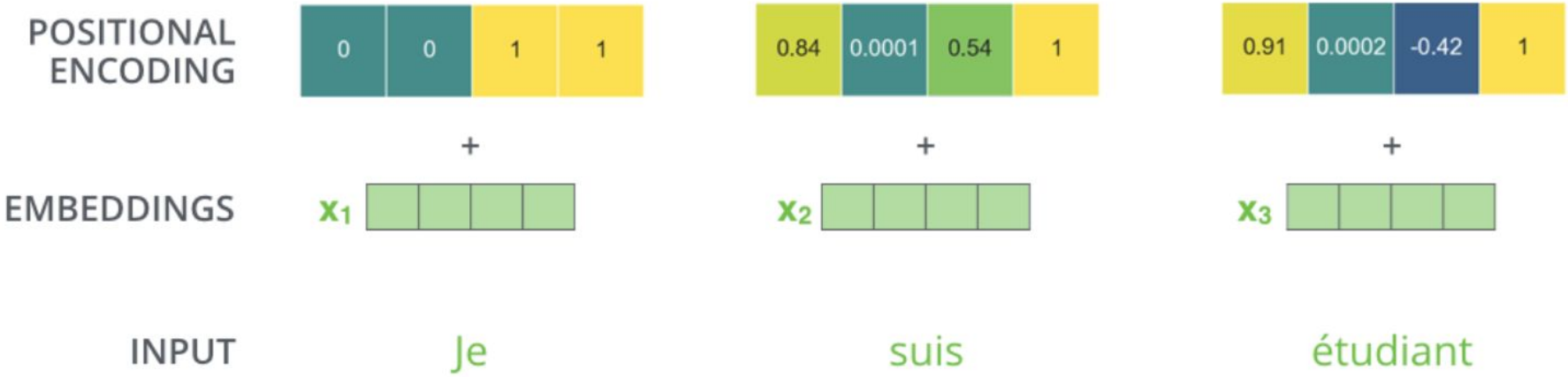
Positional Encoding





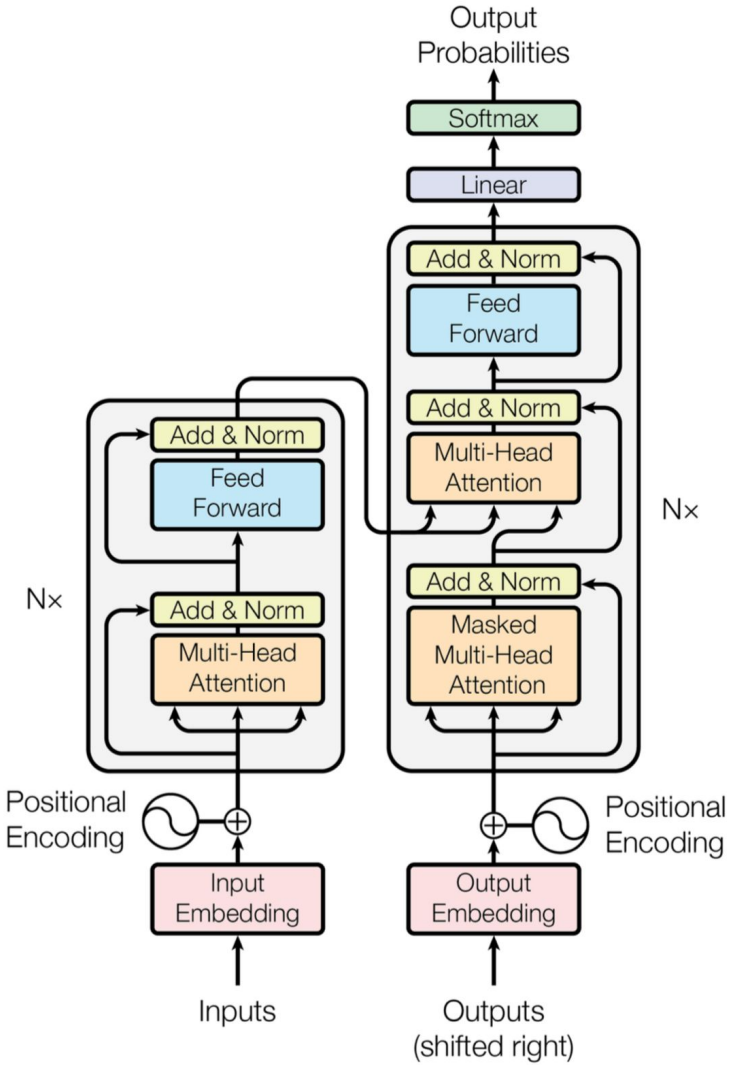
It provides meaningful distances between the embedding vectors once they're projected into Q/K/V vectors and during dot-product attention

Positional Encoding

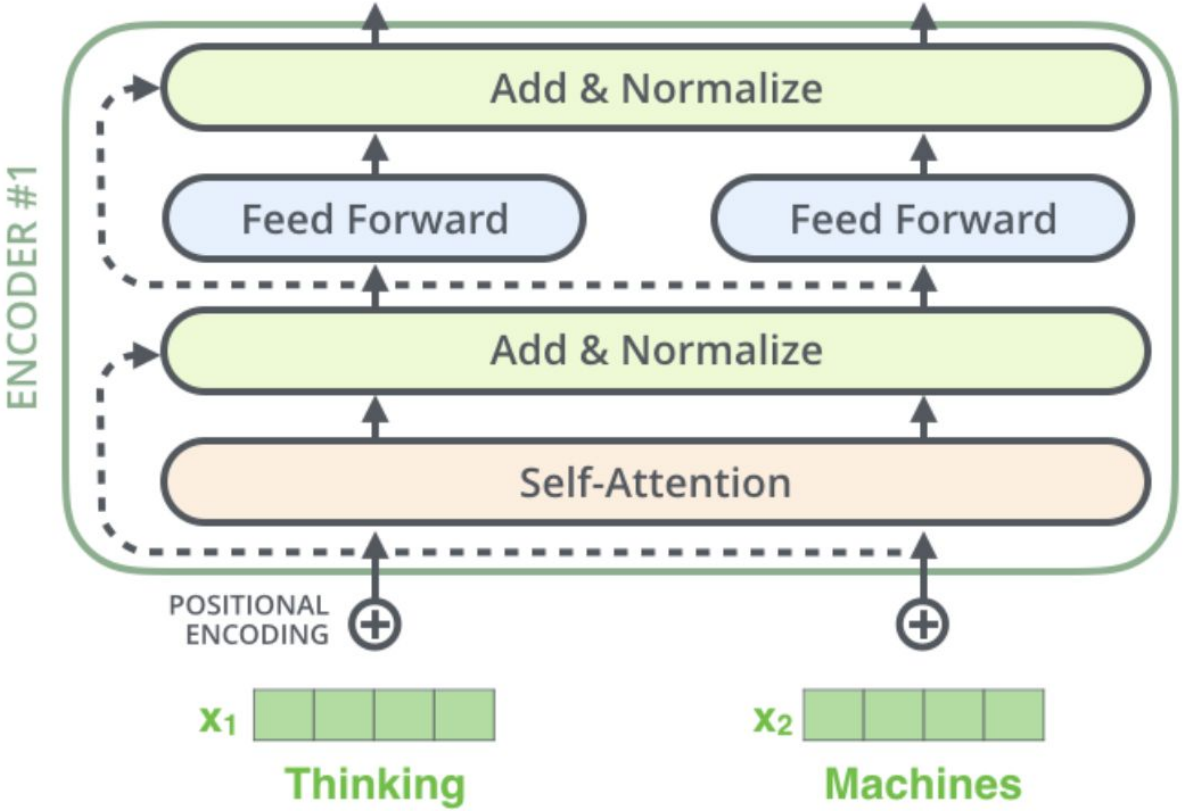


Layer Normalization

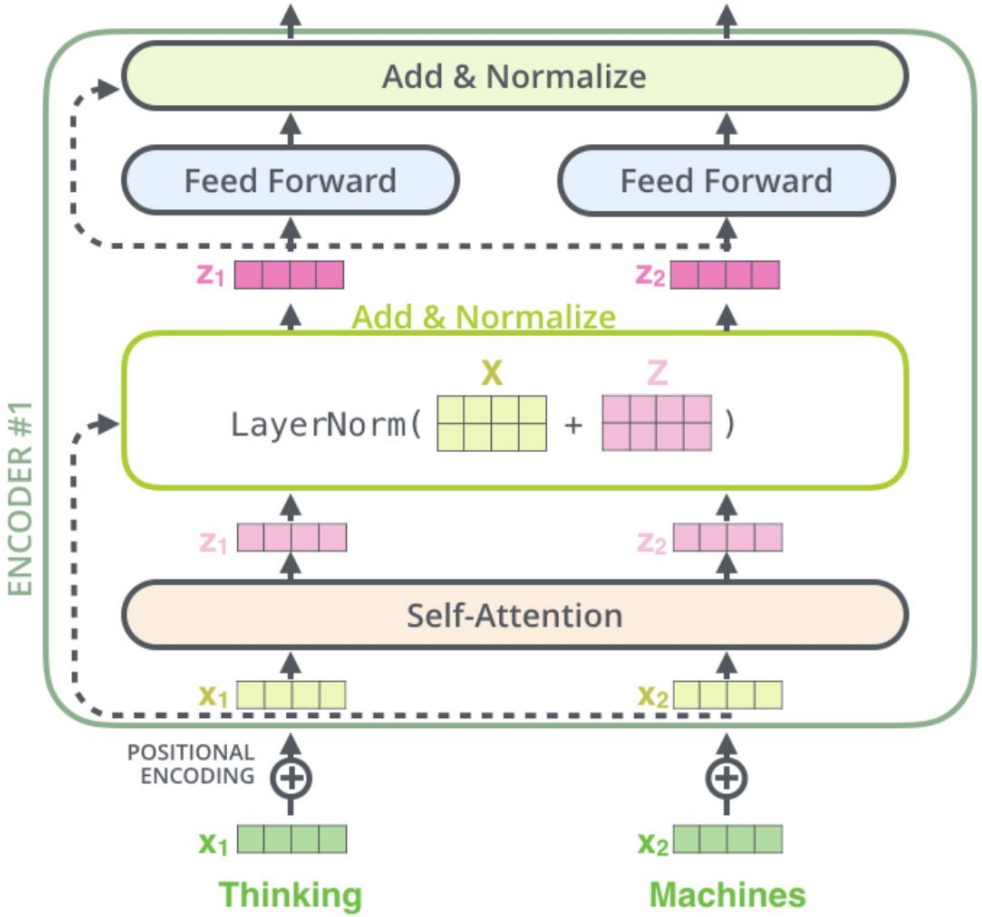
The Transformer: recap



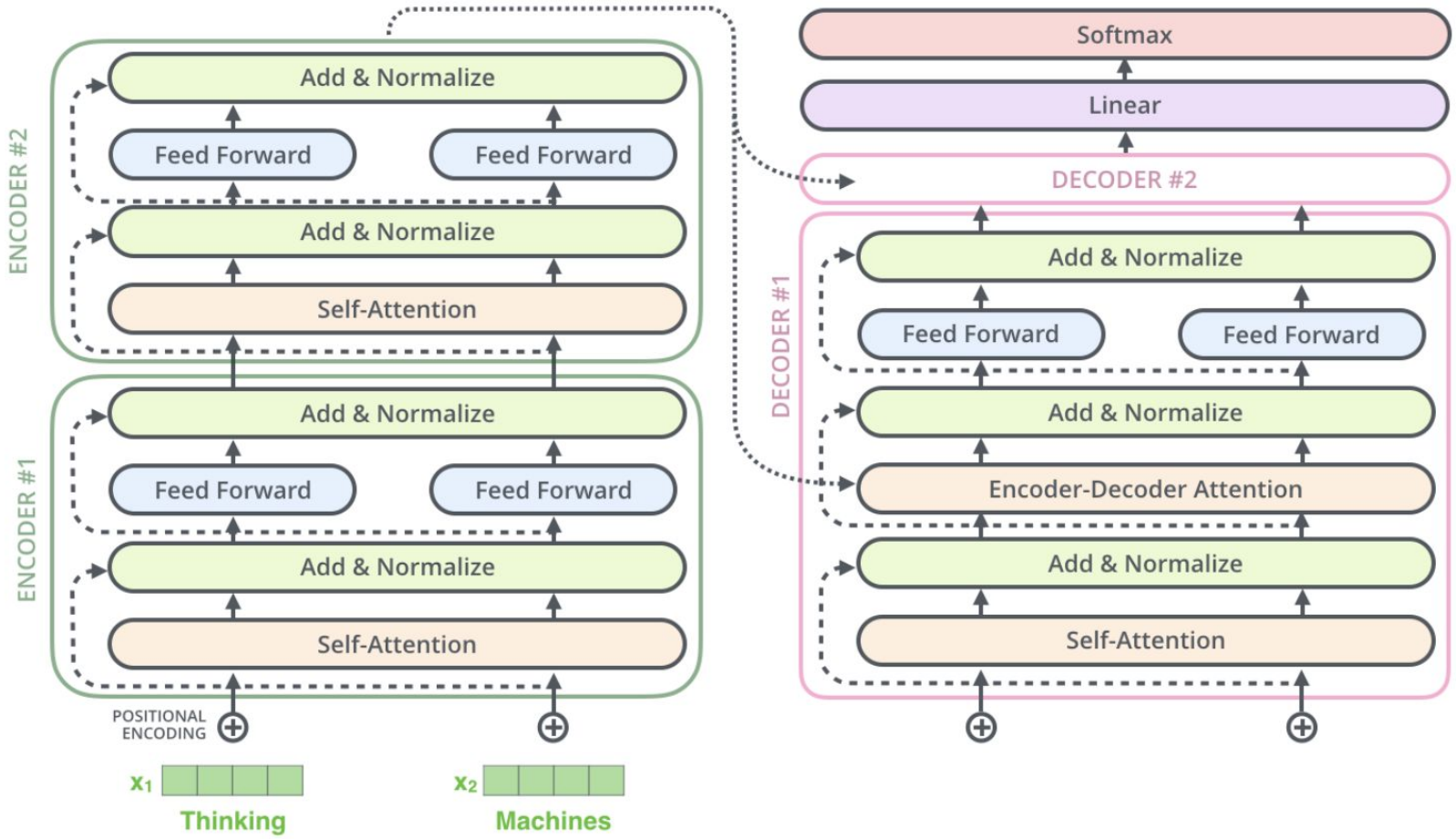
Layer Normalization



Layer Normalization



Layer Normalization

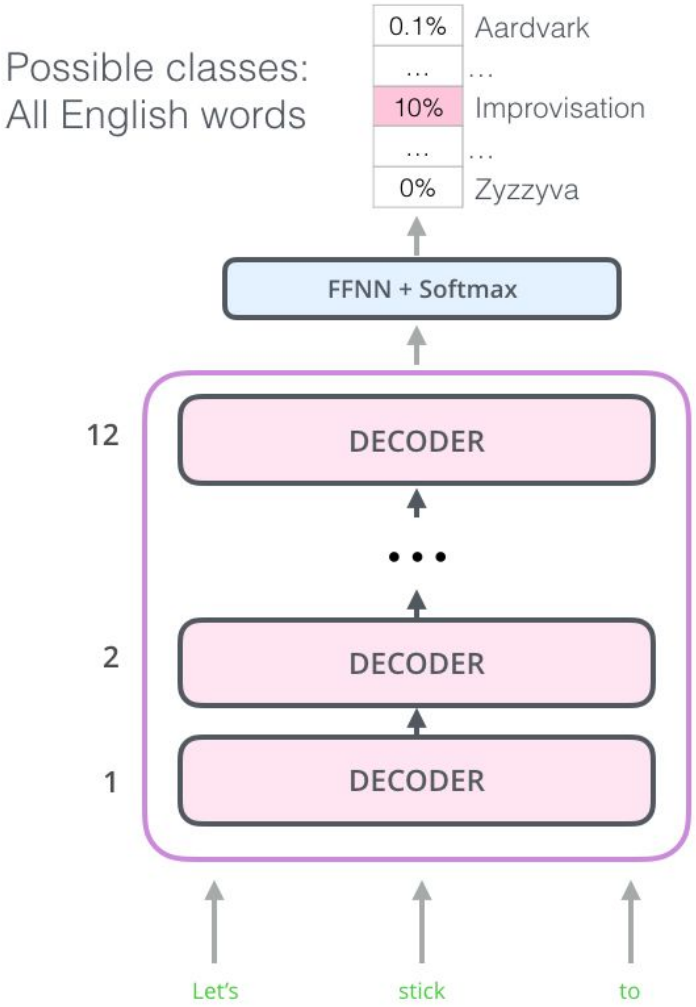


OpenAI Transformer: Pre-training Decoder for Language Modeling

OpenAI Transformer

- The Encoder-Decoder structure of the transformer made it perfect for machine translation
- But what about sentence classification?
- **<u>Main goal: pre-train a language model that can be fine-tuned for other tasks</u>**

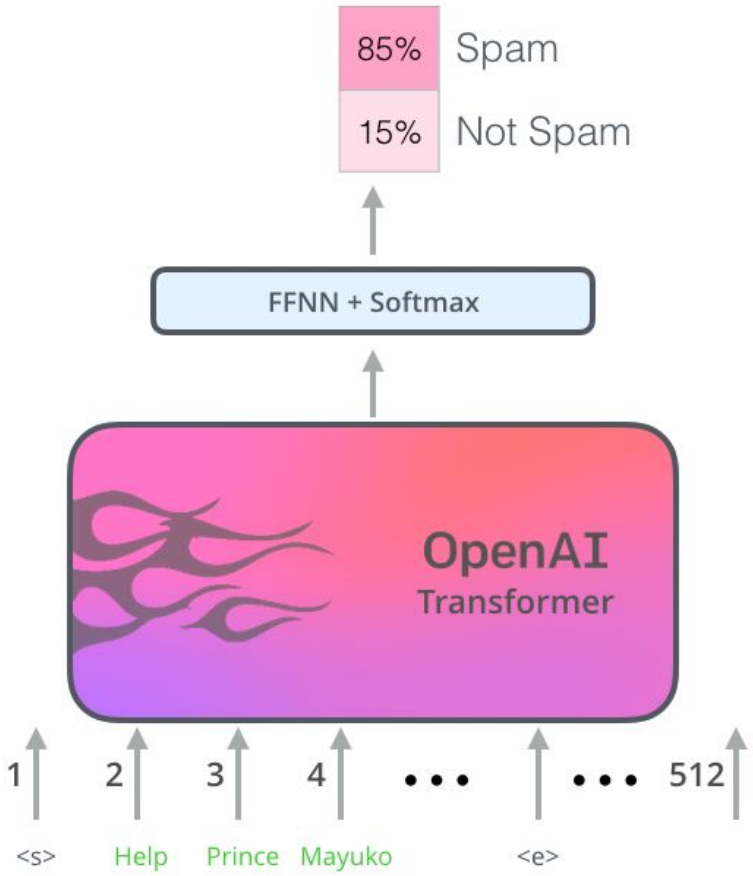
OpenAI Transformer



Differences from vanilla Transformer:

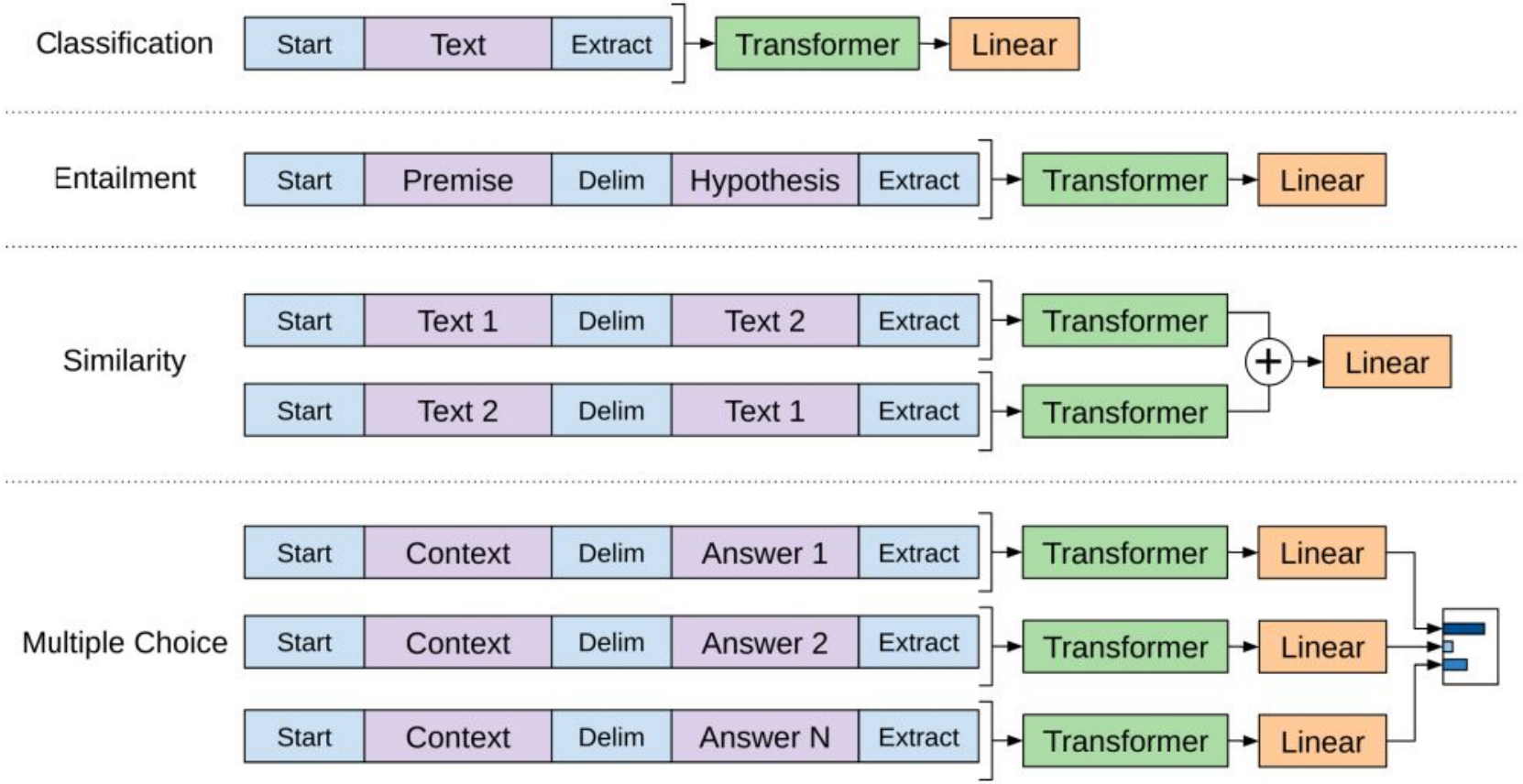
- no encoder
- decoder layers would not have the encoder-decoder attention sublayer
- Pre-train the model on predicting the next word using massive (unlabeled) datasets

OpenAI Transformer



- During pre-training phase layers have been tuned to reasonably handle language
- Now let's use it for downstream tasks (e.g. sentence classification)

Input transformations for different tasks



ELMo: context that matters

ELMo: contextualized word embeddings

"Why not give it an embedding based on the context it's used in – to both capture the word meaning in that context as well as other contextual information?"

Peters et. al., 2017, McCann et. al., 2017,
and yet again Peters et. al., 2018 in the ELMo paper

ELMo - deep contextualized word representations

ELMo

What does it stand for?

1. **E**xpedited **L**abour **M**arket **O**pinion
2. **E**lectric **L**ight **M**achine **O**rganization
3. **E**nough **L**et's **M**ove **O**n

ELMo

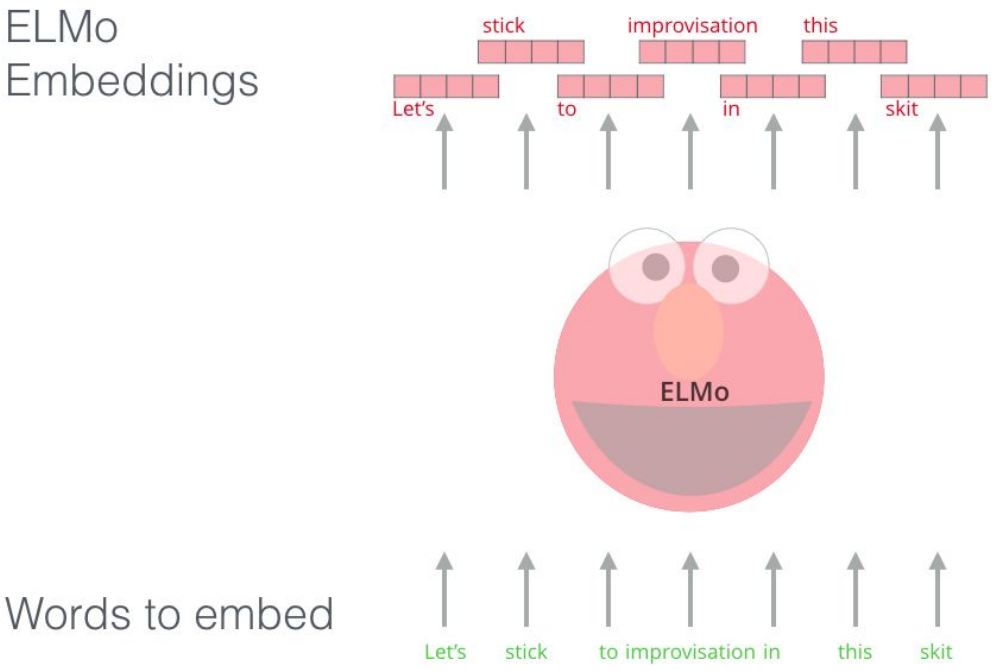
What does it stand for?

1. **E**xpedited **L**abour **M**arket **O**pinion
2. **E**lectric **L**ight **M**achine **O**rganization
3. **E**nough **L**et's **M**ove **O**n

4. **<u>Embeddings from Language Models</u>**

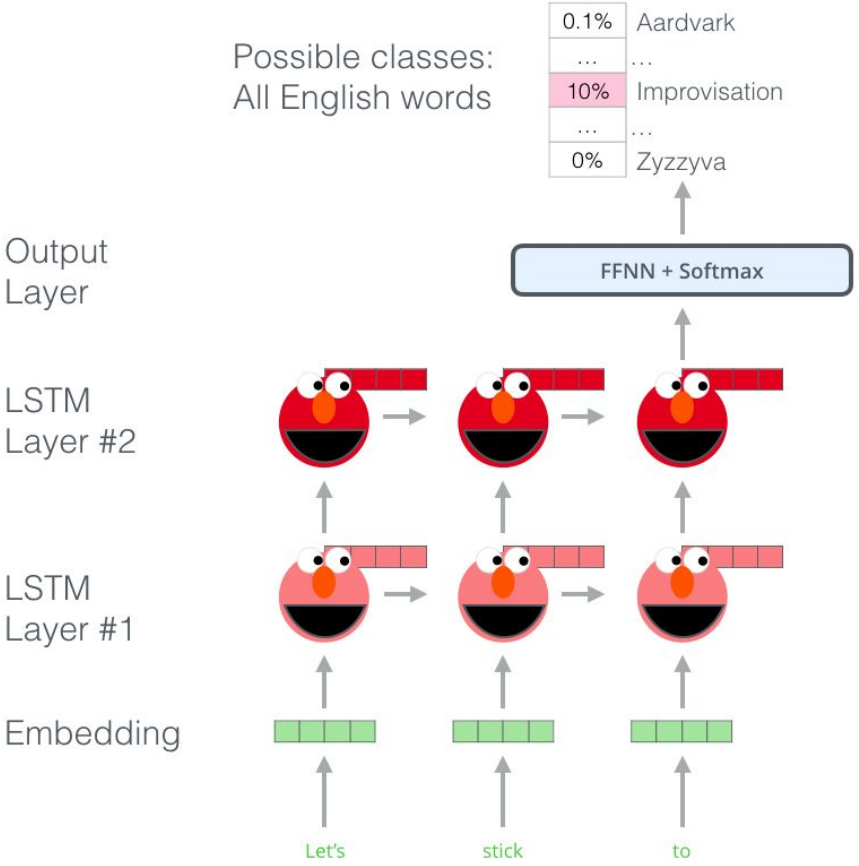
ELMo: contextualized word embeddings

ELMo: Contextualized word embeddings



- uses a bi-directional LSTM trained on Language Modeling task
- a model can learn without labels

Bidirectional Language Models (biLMs)



biLMs consist of forward and backward LMs:

- forward:

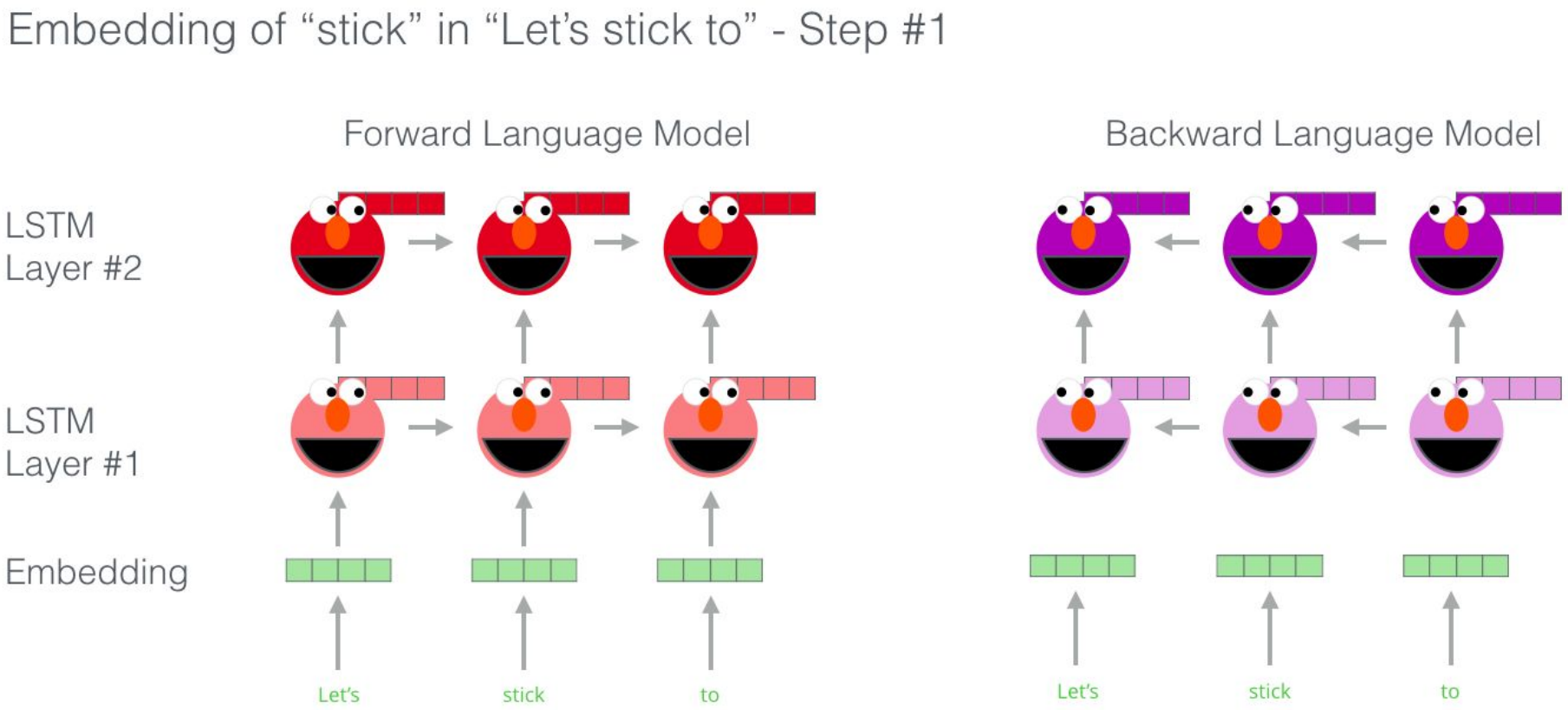
$$p(t_1,t_2,\ldots,t_N)=\prod_{k=1}^{N} p(t_k|t_1,t_2,\ldots,t_{k-1})$$

- Backward:

$$p(t_1,t_2,\ldots,t_N)=\prod_{k=1}^{N} p(t_k|t_{k+1},t_{k+2},\ldots,t_N)$$

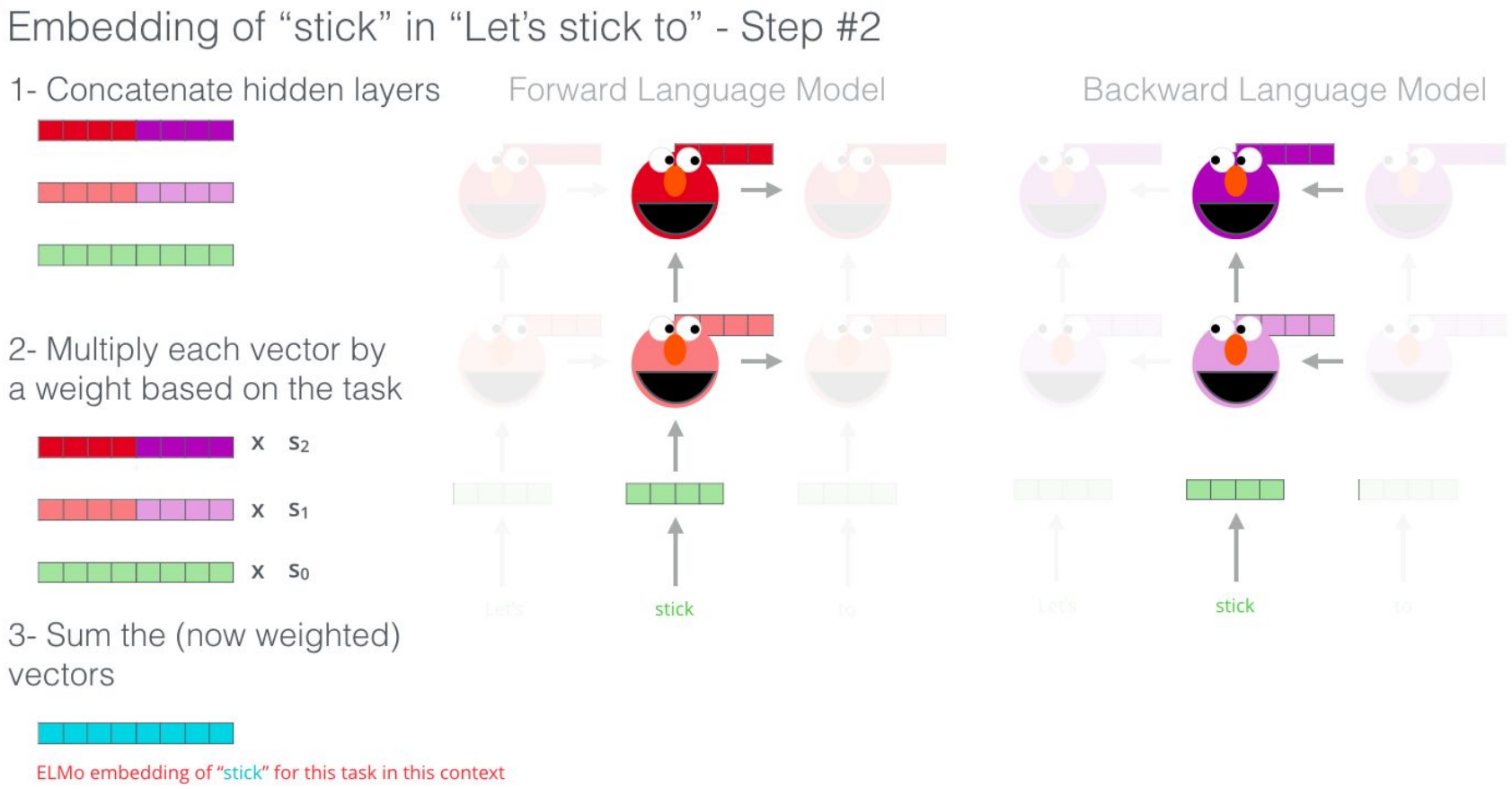
LSTM predicts next word in both directions to build biLMs

ELMo: main pipeline



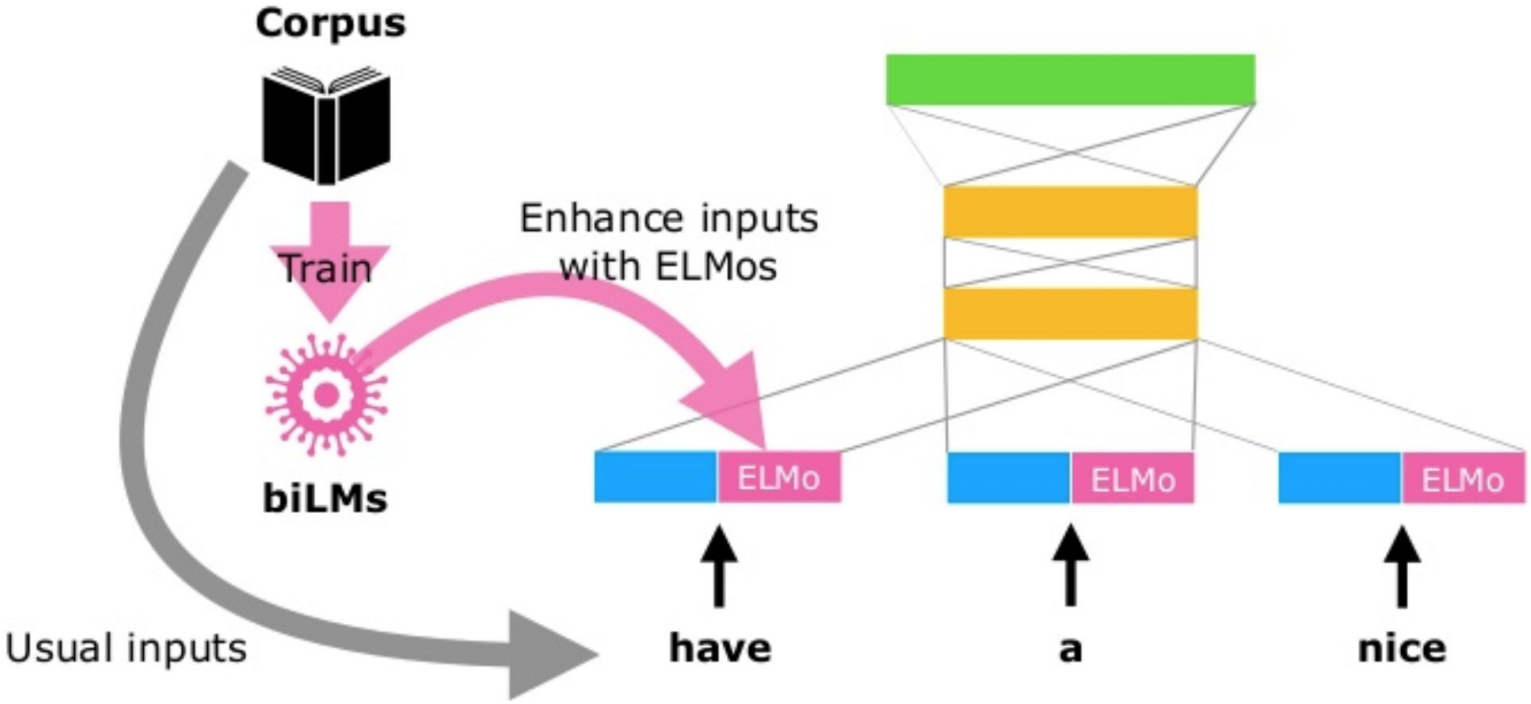
ELMo: main pipeline

ELMo represents a word as a linear combination of corresponding hidden layers:



ELMo

ELMo can be integrated to almost all neural NLP tasks with simple concatenation to the embedding layer



ELMo: overview

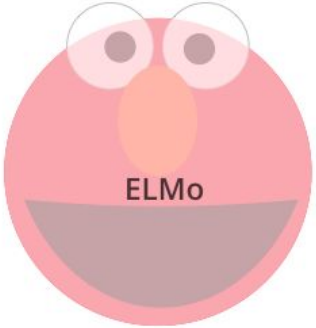
- Pretrained ELMo models: http://allennlp.org/elmo
- AllenNLP is a library on the top of PyTorch

- Higher levels seems to catch semantics while lower layer probably capture syntactic features

THE
TRANSFORMER

ULM-FiT

OpenAI
Transformer

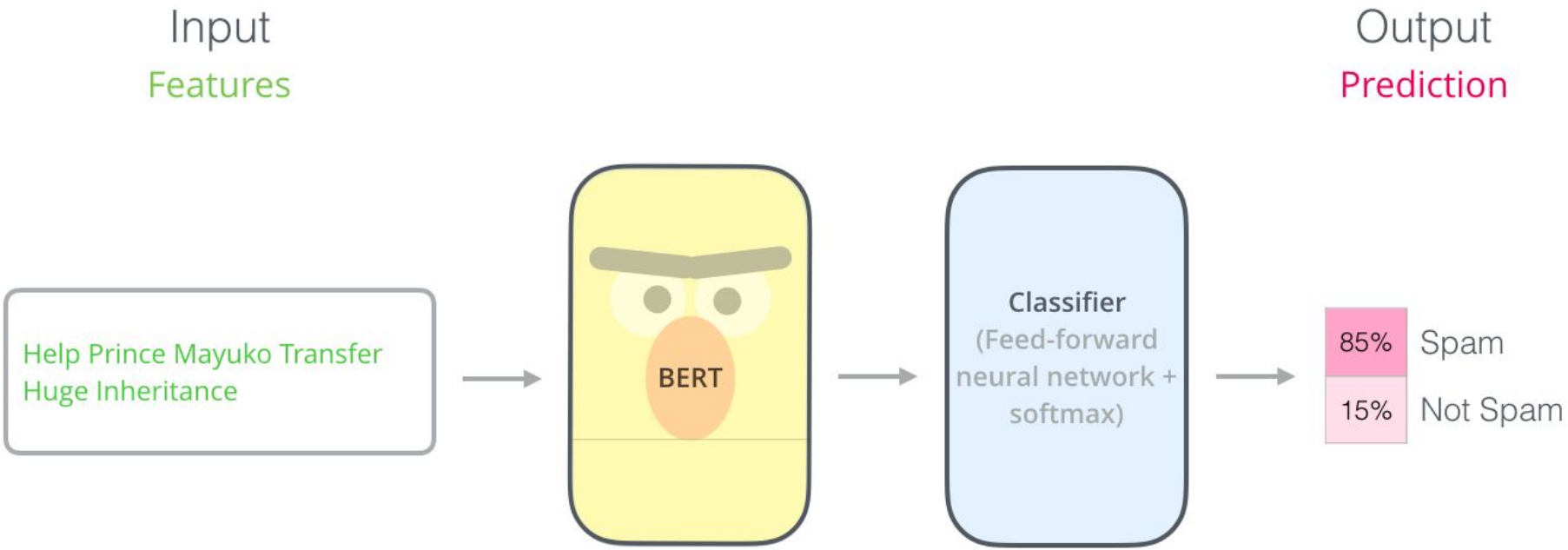

ELMo

BERT

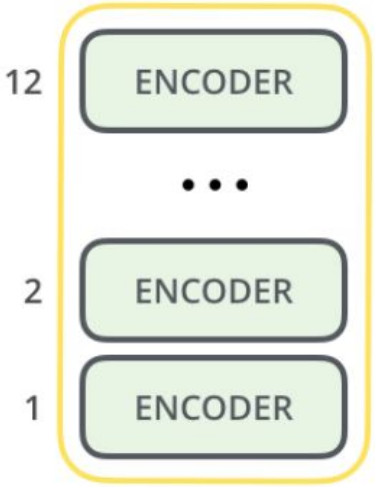
BERT

Bidirectional Encoder Representations from Transformers

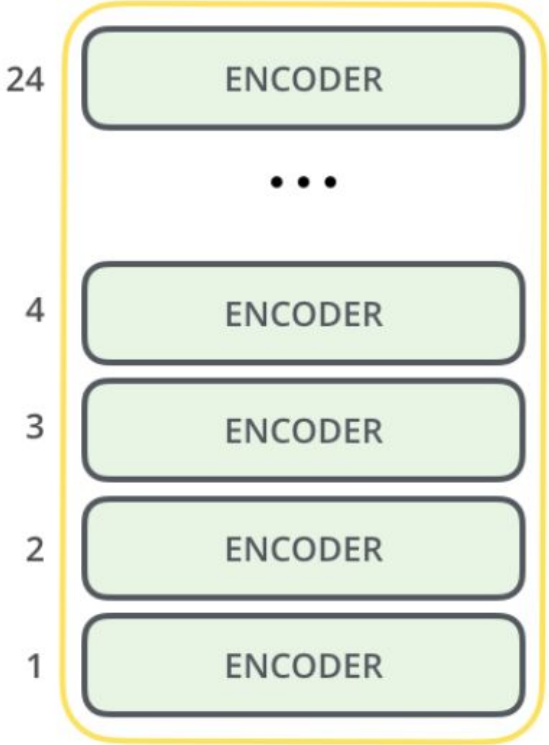
BERT


Input
Features
Help Prince Mayuko Transfer
Huge Inheritance
BERT
Classifier
(Feed-forward
neural network +
softmax)
Output
Prediction
85%
Spam
15%
Not Spam

BERT: base and large



$BERT_{BASE}$

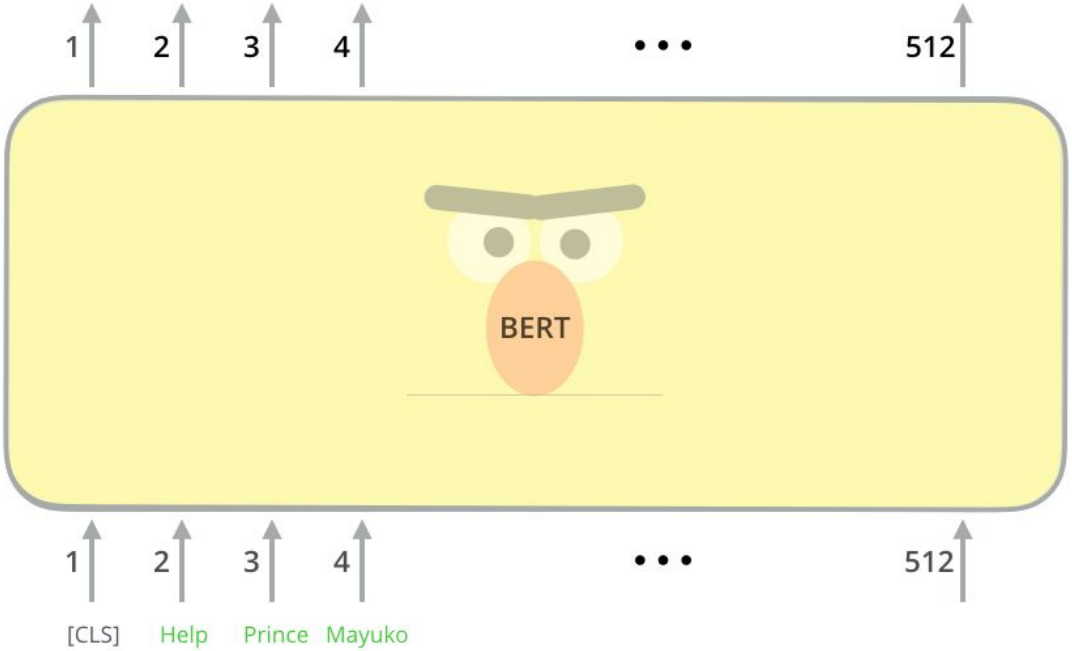


$BERT_{LARGE}$

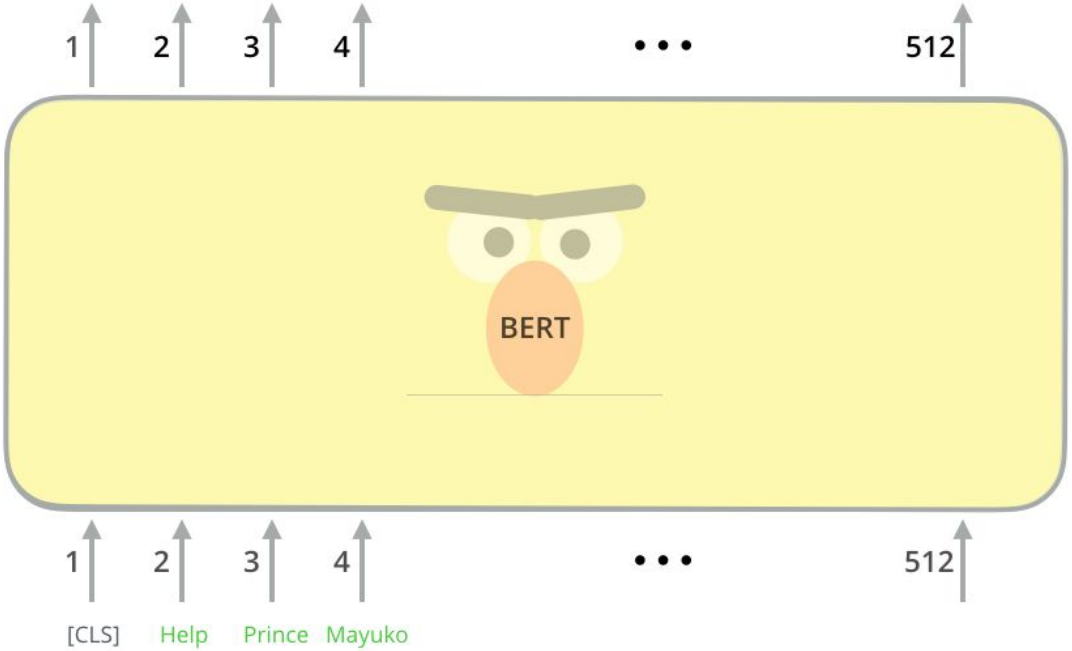
BERT vs. Transformer

		Base BERT	**Large BERT**
Encoders	6	12	24
Units in FFN	512	768	1024
Attention Heads	8	12	16

Model inputs

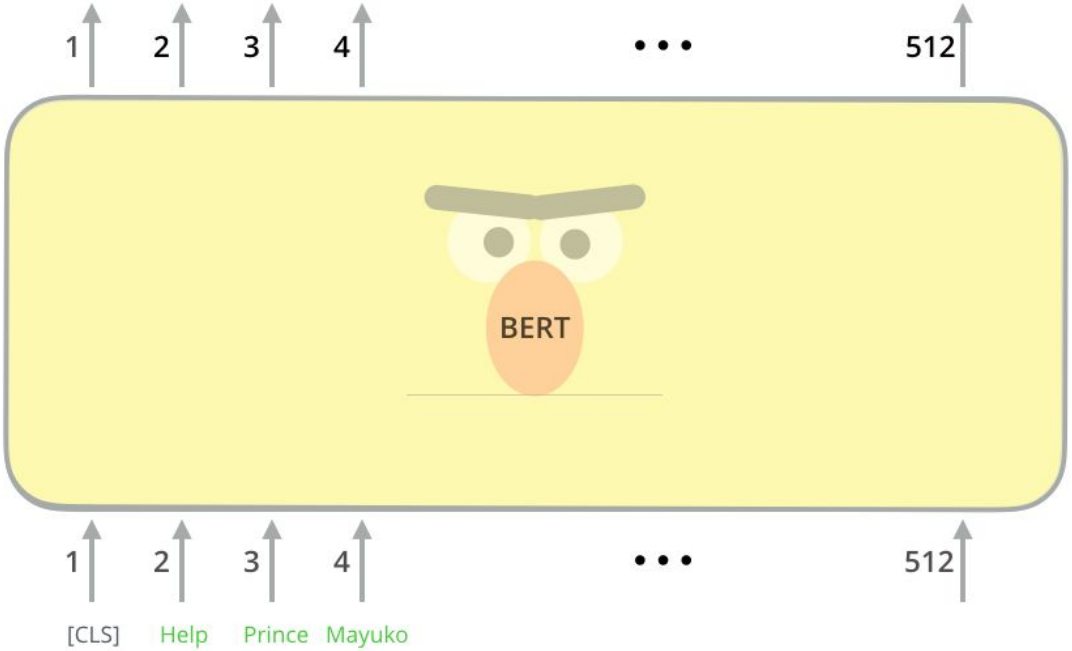


Model inputs



Identical to the Transformer up until this point

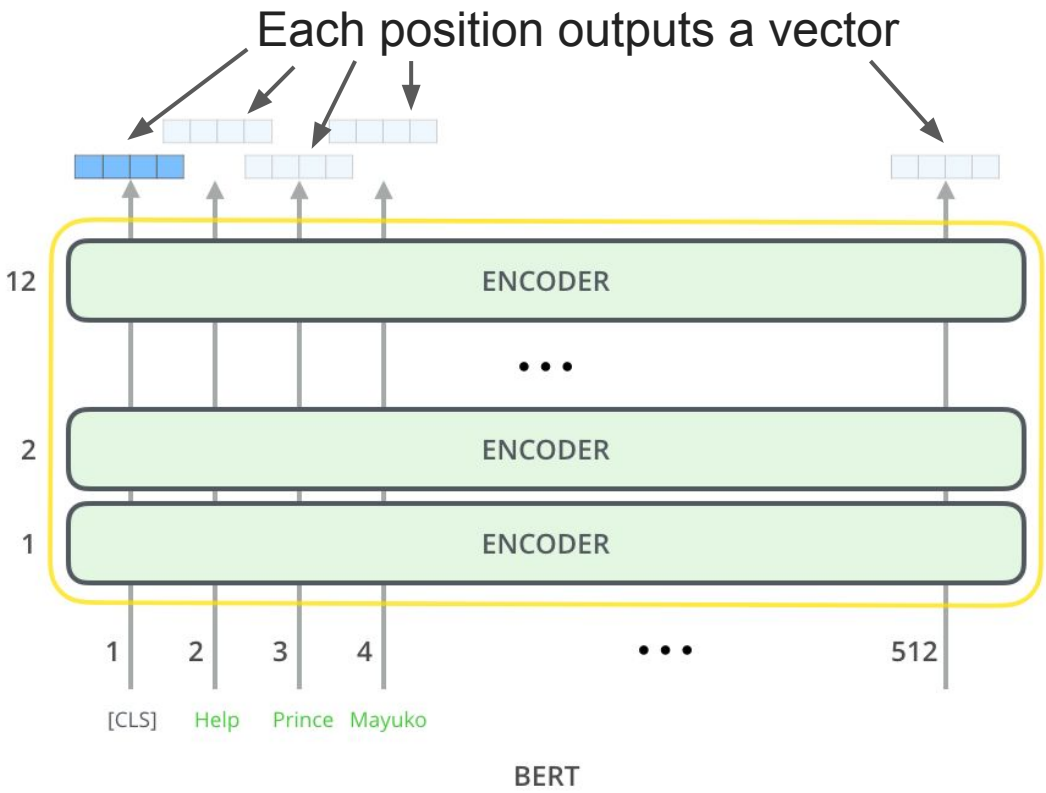
Model inputs



Identical to the Transformer up until this point

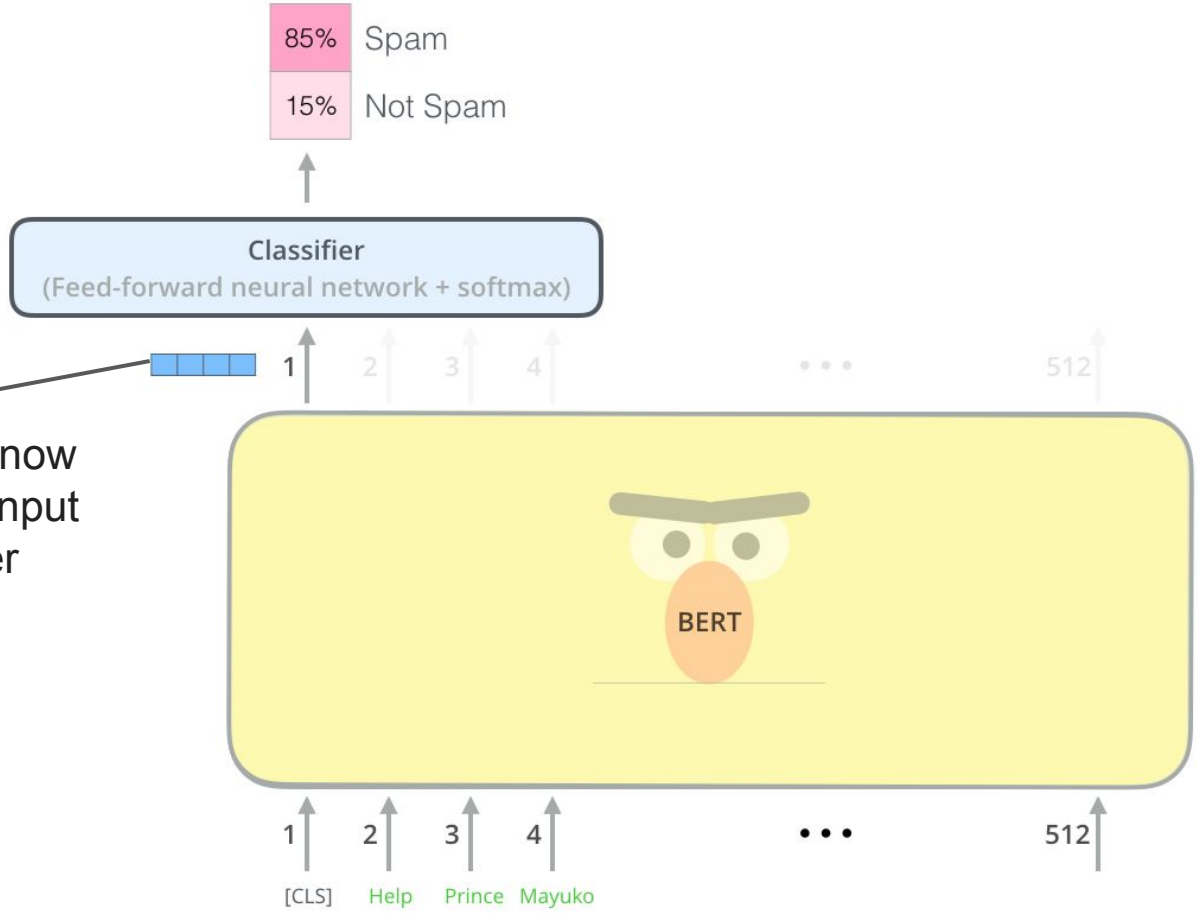
Why is BERT so special?

Model outputs



For sentence classification we focus on the first position (that we passed [CLS] token to)

Model inputs



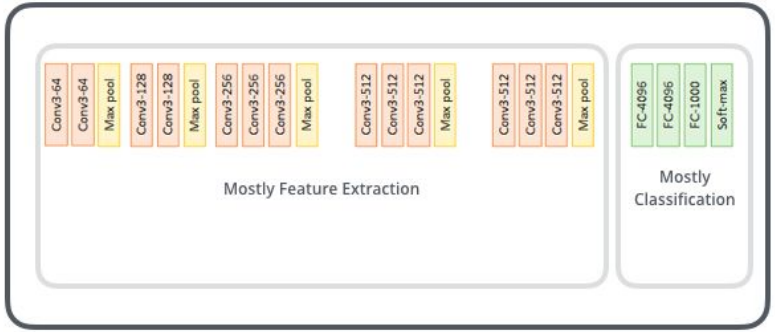
This vector can now be used as the input for a classifier

Similar to CNN concept!

Input

Features

VGG-16



Output

Prediction

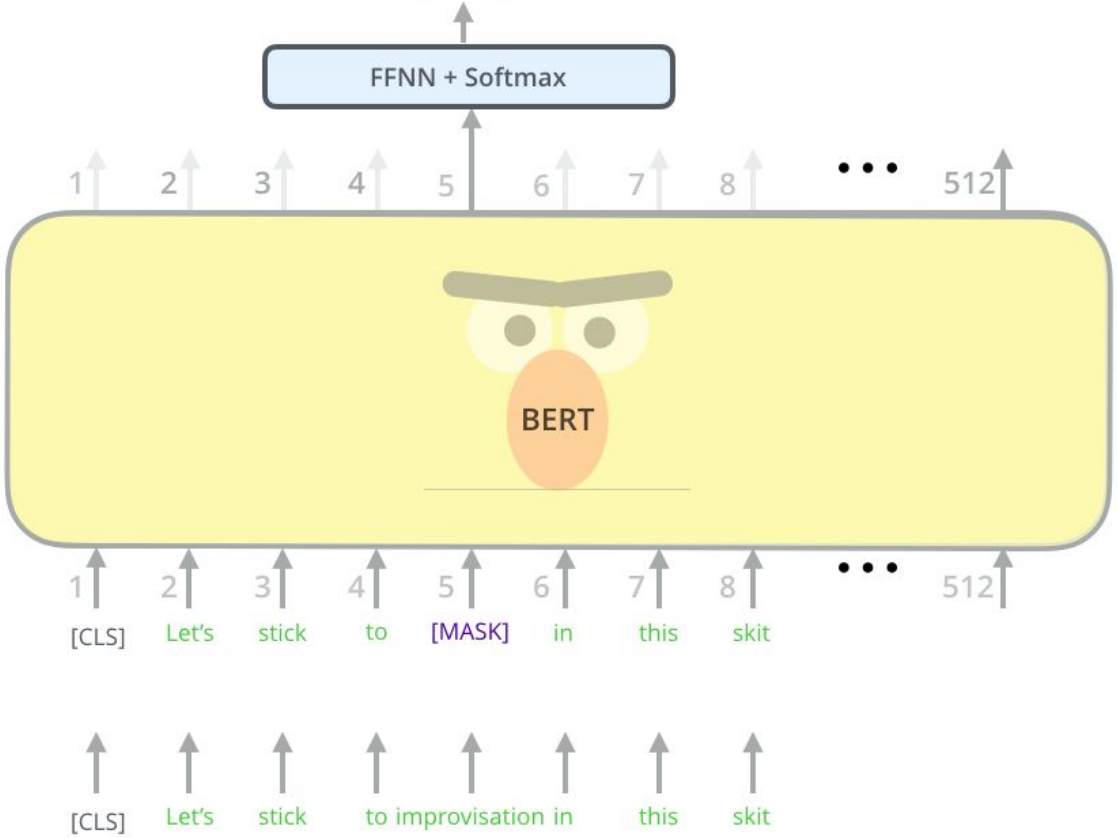
0.2%	Kit fox
0.1%	English setter
95%	Egyptian cat
1%	Great Dane
	…
0%	Hotdog

BERT: pre-training

Use the output of the masked word's position to predict the masked word

Possible classes: All English words

0.1%	Aardvark
…	…
10%	Improvisation
…	…
0%	Zyzzyva



Randomly mask 15% of tokens

Input

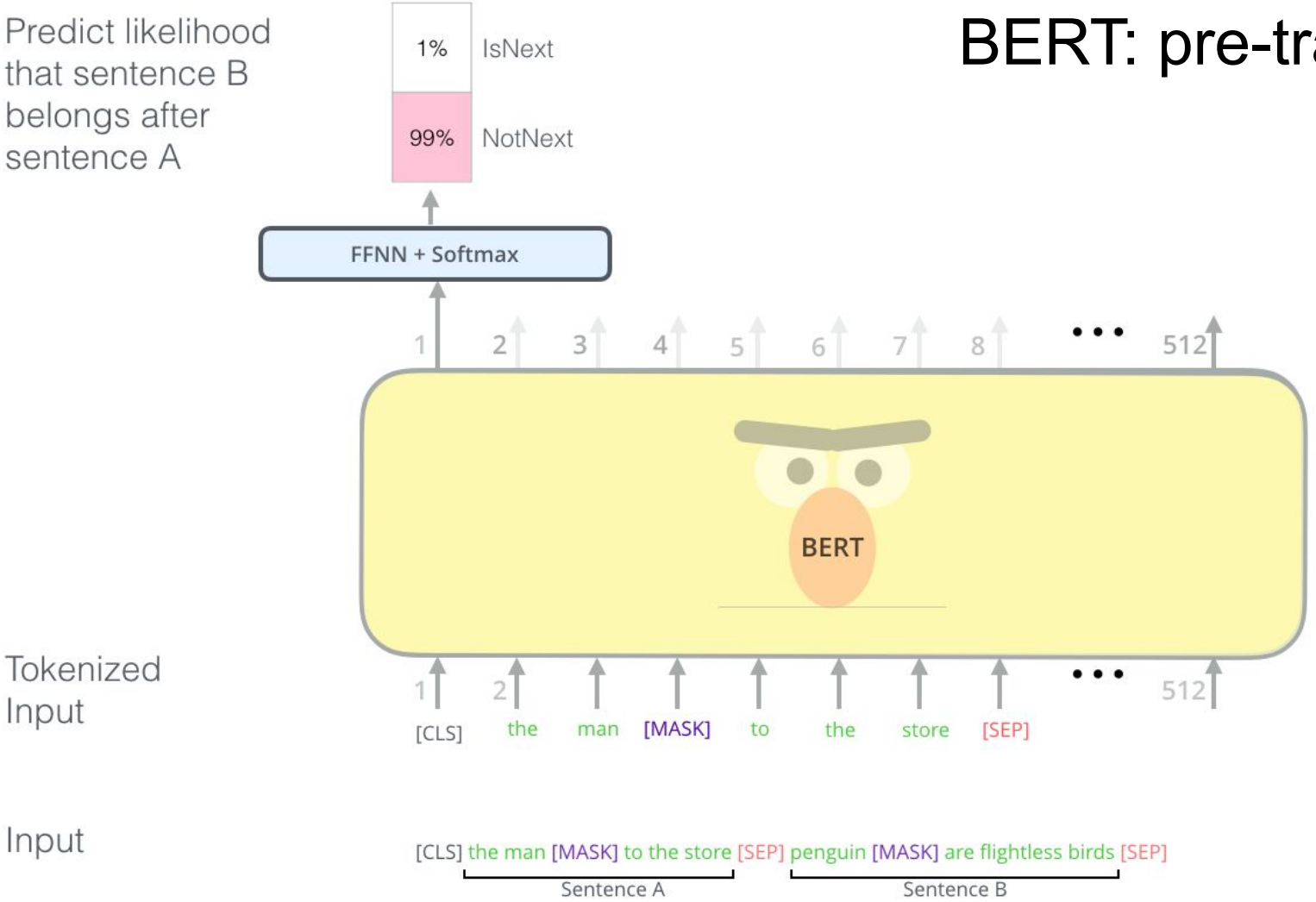
BERT: pre-training

- “Masked Language Model” approach

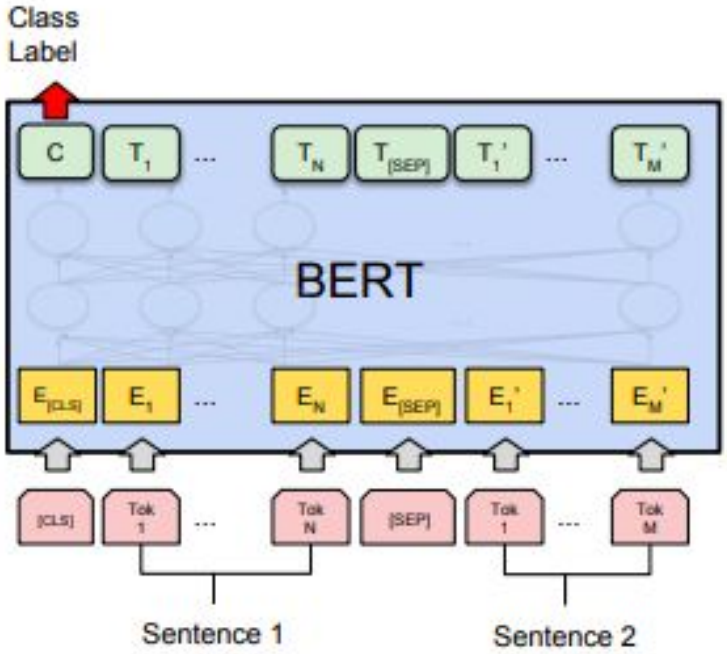
- To make BERT better at handling relationships between multiple sentences, the pre-training process includes an additional task:

  “Given two sentences (A and B),is B likely to be the sentence that follows A, or not?”

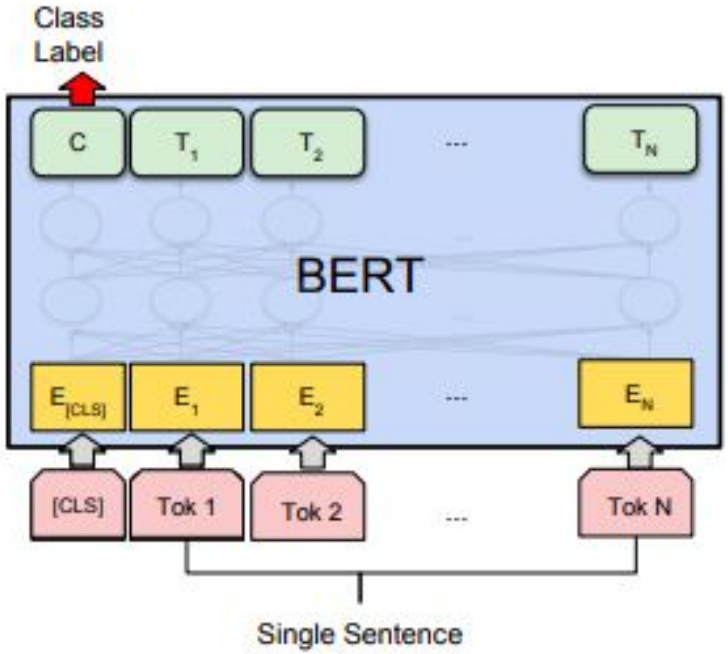
BERT: pre-training



BERT: fine-tuning for different tasks

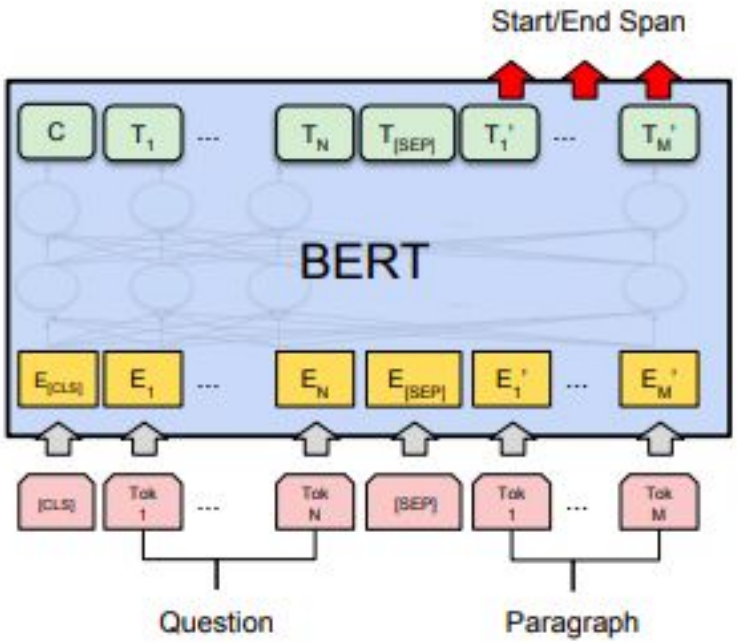


(a) Sentence Pair Classification Tasks: MNLI, QQP, QNLI, STS-B, MRPC, RTE, SWAG

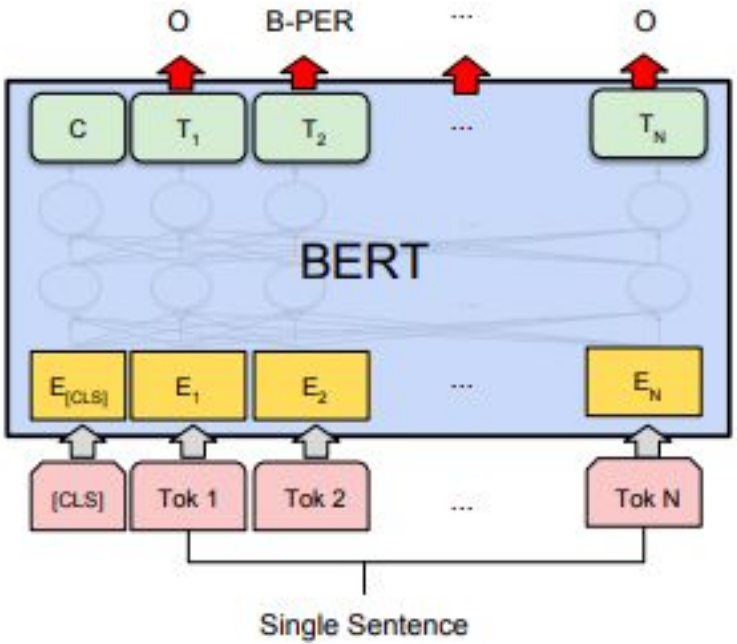


(b) Single Sentence Classification Tasks: SST-2, CoLA

BERT: fine-tuning for different tasks

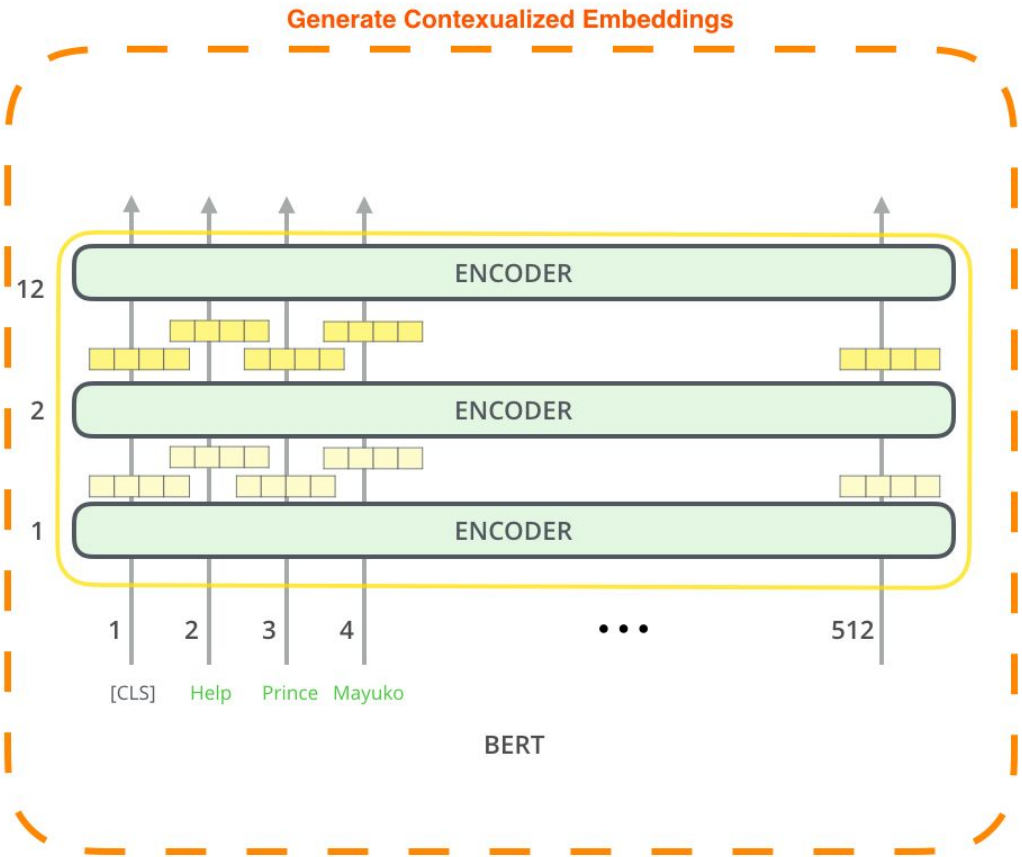


(c) Question Answering Tasks:
SQuAD v1.1

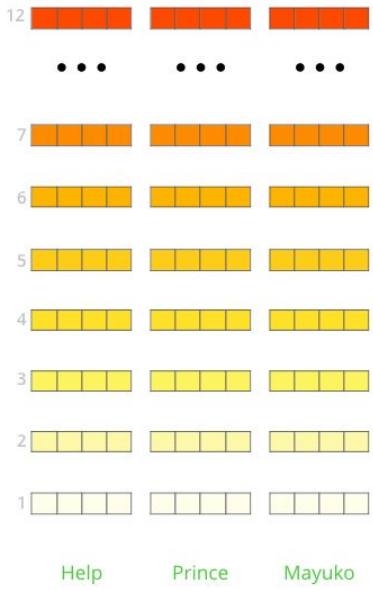


(d) Single Sentence Tagging Tasks:
CoNLL-2003 NER

BERT for feature extraction



The output of each encoder layer along each token's path can be used as a feature representing that token.

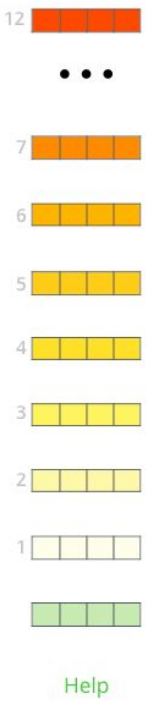


But which one should we use?

BERT for feature extraction

What is the best contextualized embedding for "Help" in that context?

For named-entity recognition task CoNLL-2003 NER



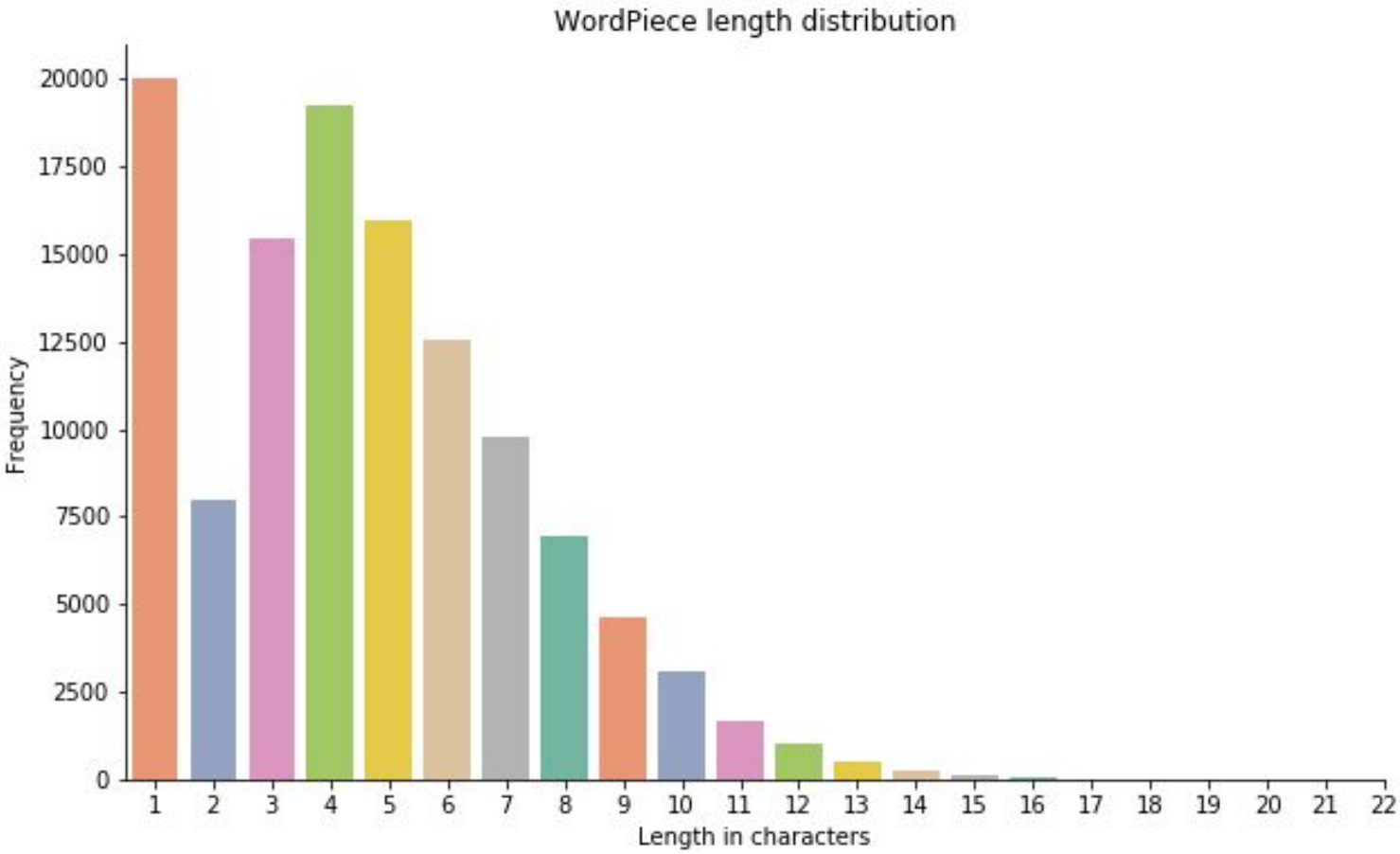
		Dev F1 Score
First Layer	Embedding	91.0
Last Hidden Layer	12	94.9
Sum All 12 Layers	12 + ... + 2 + 1 =	95.5
Second-to-Last Hidden Layer	11	95.6
Sum Last Four Hidden	12 + 11 + 10 + 9 =	95.9
Concat Last Four Hidden	9 10 11 12	96.1

BERT: tokenization

<u>Example:</u> Unaffable -> un, ##aff, ##able

- Single model for 104 languages with a large shared vocabulary (119,547 WordPiece model)
- Non-word-initial units are prefixed with ##
- The first 106 symbols: constants like PAD and UNK
- 36.5% of the vocabulary are non-initial word pieces
- The alphabet consists of 9,997 unique characters that are defined as word-initial (C) and continuation symbols (##C), which together make up 19,994 word pieces
- The rest are multicharacter word pieces of various length.

BERT: tokenization



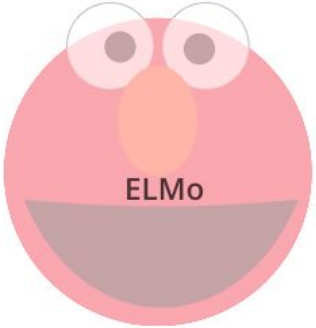
BERT: overview

- BERT repo
- Try out BERT on TPU
- WordPieces Tokenizer
- PyTorch Implementation of BERT

THE
TRANSFORMER

ULM-FiT

OpenAI
Transformer


ELMo

BERT